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1. Overview & Conventions

1.1 Base URL & Versioning

All endpoints are served under a versioned base path. The current production version is v1. When a breaking change is introduced, a new version prefix will be published alongside the previous one for a deprecation window of no less than six months.

```
https://api.agrisync.com/v1
```

Staging and development environments follow the same structure:

```
https://staging-api.agrisync.com/v1
```

```
https://dev-api.agrisync.com/v1
```

1.2 Authentication Flow

The API uses JSON Web Tokens (JWT) for stateless authentication. After a successful login, the server returns an access token (short-lived, 30 minutes) and a refresh token (long-lived, 7 days). The access token must be included as a Bearer token in the Authorization header of every subsequent request.

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIs...
```

When the access token expires, the client must call the refresh endpoint with the refresh token to obtain a new pair. The previous refresh token is invalidated on use (rotation). If the refresh token itself has expired, the user must log in again.

1.3 Request & Response Format

All request and response bodies use JSON with UTF-8 encoding. Dates follow ISO 8601 format (YYYY-MM-DD for dates, YYYY-MM-DDTHH:mm:ssZ for timestamps). Monetary and quantity values are transmitted as strings with up to four decimal places to avoid floating-point rounding during transport. UUIDs are used as primary identifiers throughout.

Successful responses follow this envelope:

```
{ "data": { ... }, "meta": { "request_id": "a1b2c3", "timestamp": "2025-..." } }
```

List endpoints include pagination metadata:

```
{ "data": [ ... ], "meta": { "current_page": 1, "per_page": 25, "total": 142, "last_page": 6 } }
```

1.4 Error Handling

Error responses use standard HTTP status codes and include a machine-readable error code alongside a human-readable message. Validation errors enumerate each failing field.

HTTP	Code	When
400	bad_request	Malformed JSON or missing required fields
401	unauthenticated	Missing, expired, or invalid token
403	forbidden	Authenticated but insufficient role privileges

HTTP	Code	When
404	not_found	Resource does not exist or has been soft-deleted
409	conflict	Duplicate entry (e.g. duplicate client_id in postings)
422	validation_error	Input fails validation; see errors object
429	rate_limited	Too many requests; retry after Retry-After header
500	server_error	Unexpected failure; report request_id to support

Validation error example:

```
{ "error": { "code": "validation_error", "message": "The given data was invalid.", "errors":
{ "email": ["The email field must be a valid email address."], "password": ["The password
field must be at least 8 characters."] } } }
```

1.5 Pagination

All list endpoints accept page and per_page query parameters. The default page size is 25, maximum is 100. Results are ordered by created_at DESC unless the endpoint specifies otherwise.

1.6 Rate Limiting

The API enforces 120 requests per minute per authentication token. Rate limit status is communicated via response headers: X-RateLimit-Limit, X-RateLimit-Remaining, and Retry-After (seconds, present only when throttled).

2. Authentication

POST /auth/login

Authenticates a user and returns a JWT token pair. The server logs the event and updates last_login_at on the user record.

Auth: None (public) | **Roles:** Any | **Success:** 200 OK

Request Body (JSON)

```
{
  "email": "ahmed@agrisync.com",
  "password": "s3cureP@ss"
}
```

Response Body

```
{
  "data": {
    "access_token": "eyJhbG...",
    "refresh_token": "dGhpcyBp...",
    "token_type": "Bearer",
    "expires_in": 1800,
    "user": {
      "id": "9c7f...",
      "name": "Ahmed Benali",
      "email": "ahmed@agrisync.com",
      "roles": ["technician"],
      "preferred_lang": "fr"
    }
  }
}
```

Error Responses

Code	Description
401	Invalid credentials
403	Account deactivated

Note: Failed login attempts are logged. After 5 consecutive failures within 15 minutes, the account is temporarily locked for 30 minutes.

POST /auth/refresh

Exchanges a valid refresh token for a new token pair. The previous refresh token is invalidated immediately (single-use rotation).

Auth: Refresh token | **Success:** 200 OK

Request Body (JSON)

```
{
  "refresh_token": "dGhpcyBp..."
}
```

Response Body

```
{
  "data": {
    "access_token": "eyJhbG... (new)",
    "refresh_token": "bmV3IHJl... (new)",
    "token_type": "Bearer",
    "expires_in": 1800
  }
}
```

Error Responses

Code	Description
401	Refresh token expired or already used

POST /auth/logout

Invalidates the current token pair. The refresh token is added to a server-side deny list.

Auth: Bearer token | **Success:** 200 OK

Request Body (JSON)

```
{
  "refresh_token": "dGhpcyBp..."
}
```

Response Body

```
{
  "data": { "message": "Logged out successfully." }
}
```

POST /auth/password/forgot

Sends a password reset link to the given email. The endpoint always returns 200 regardless of whether the email exists, to prevent user enumeration.

Auth: None (public) | **Success:** 200 OK

Request Body (JSON)

```
{
  "email": "ahmed@agrisync.com"
}
```

Response Body

```
{
  "data": { "message": "If an account with that email exists, a reset link has been sent." }
}
```

POST /auth/password/reset

Resets the password using a token received via email. The token is single-use and expires after 60 minutes.

Auth: None (public) | **Success:** 200 OK

Request Body (JSON)

```
{
  "token": "a1b2c3d4e5...",
  "password": "newS3cureP@ss",
  "password_confirmation": "newS3cureP@ss"
}
```

Response Body

```
{
  "data": { "message": "Password has been reset." }
}
```

Error Responses

Code	Description
400	Token invalid or expired
422	Password does not meet requirements

3. User Management

User accounts are created exclusively by administrators. There is no self-registration. Users are soft-deactivated rather than deleted to preserve audit trail integrity.

GET /users

Returns a paginated list of all users. Supports filtering by role, active status, and search by name or email.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
role	string	No	Filter: technician, manager, admin
is_active	boolean	No	Filter by account status
search	string	No	Partial match on name or email
page	integer	No	Page number (default 1)
per_page	integer	No	Results per page (default 25, max 100)

Response Body

```
{
  "data": [
    {
      "id": "9c7f...",
      "name": "Ahmed Benali",
      "email": "ahmed@agrisync.com",
      "roles": ["technician"],
      "is_active": true,
      "preferred_lang": "fr",
      "last_login_at": "2025-03-10T08:45:00Z",
      "created_at": "2025-01-15T10:00:00Z"
    }
  ],
  "meta": { "current_page": 1, "per_page": 25, "total": 12, "last_page": 1 }
}
```

POST /users

Creates a new user account and assigns one or more roles. A welcome email is sent to the user with temporary credentials.

Auth: Bearer token | **Roles:** admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "name": "Fatima Zahra",
  "email": "fatima@agrisync.com",
  "password": "Temp1234!",
}
```

```
"roles": ["manager"],
"preferred_lang": "fr"
}
```

Response Body

```
{
  "data": {
    "id": "b3e1...",
    "name": "Fatima Zahra",
    "email": "fatima@agrisync.com",
    "roles": ["manager"],
    "is_active": true,
    "preferred_lang": "fr",
    "created_at": "2025-04-12T14:30:00Z"
  }
}
```

Error Responses

Code	Description
409	Email already registered
422	Validation failed

PUT /users/{id}

Updates a user's profile information. Role changes take effect immediately. Password changes invalidate all existing tokens for that user.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Request Body (JSON)

```
{
  "name": "Fatima Zahra Idrissi",
  "roles": ["manager", "technician"],
  "preferred_lang": "en"
}
```

Response Body

```
{
  "data": { "id": "b3e1...", "name": "Fatima Zahra Idrissi", ... }
}
```

Error Responses

Code	Description
404	User not found

PATCH /users/{id}/deactivate

Suspends the user account. All active tokens are revoked. The user cannot log in until reactivated. Historical data and audit logs referencing this user are preserved.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Response Body

```
{
  "data": { "id": "b3e1...", "is_active": false, "updated_at": "..."}
}
```

Note: Deactivating the last admin account is rejected with a 409 error.

PATCH /users/{id}/activate

Restores a previously deactivated account.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Response Body

```
{
  "data": { "id": "b3e1...", "is_active": true, "updated_at": "..."}
}
```

4. Plot Management

Plots are the fundamental organisational unit. Every operation and report is scoped to a plot. The surface area drives per-hectare calculations in reporting, and the season start date determines cumulative period boundaries.

GET /plots

Lists all plots with optional filtering by crop type and active status. Includes the current season's summary statistics when `include=stats` is passed.

Auth: Bearer token | **Roles:** technician, manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
crop_type	string	No	Filter by crop category
is_active	boolean	No	Default: true
include	string	No	Comma-separated: stats, custom_fields
page	integer	No	Page number

Response Body

```
{
  "data": [
    {
      "id": "d4a2...",
      "name": "Bloc A - Clementines",
      "surface_area_ha": "12.5000",
      "crop_type": "Citrus",
      "variety": "Nour",
      "season_start_date": "2024-10-01",
      "is_active": true,
      "created_at": "2024-09-15T10:00:00Z"
    }
  ],
  "meta": { ... }
}
```

POST /plots

Registers a new plot. The surface area must be a positive decimal. Season start date is required for cumulative calculations.

Auth: Bearer token | **Roles:** admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "name": "Bloc B - Olives",
  "surface_area_ha": "8.7500",
  "crop_type": "olive",
  "season_start_date": "2024-10-01"
}
```

```
"variety": "Picholine",
"season_start_date": "2024-11-01"
}
```

Error Responses

Code	Description
422	surface_area_ha must be greater than 0

PUT /plots/{id}

Updates plot attributes. Changing surface_area_ha does not retroactively alter historical per-hectare calculations; those are computed at query time against the current value.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Request Body (JSON)

```
{
  "name": "Bloc B - Olives (Extended)",
  "surface_area_ha": "10.2500",
  "variety": "Picholine Marocaine"
}
```

DELETE /plots/{id}

Soft-deletes a plot by setting is_active = false. Plots with existing operation records cannot be hard-deleted. The plot no longer appears in selection lists but remains available in historical reports.

Auth: Bearer token | **Roles:** admin | **Success:** 204 No Content

Note: Returns 409 if attempting to delete a plot with linked operations and no force flag.

5. Water Configuration

The water configuration defines the measurement unit used across all irrigation entries. Pricing is managed separately through the Price History system (Section 9), allowing price changes without disrupting historical cost records.

GET /config/water

Returns the current water configuration, including the active price resolved from price history.

Auth: Bearer token | **Roles:** manager (read-only), admin | **Success:** 200 OK

Response Body

```
{
  "data": {
    "id": "e5f6...",
    "unit": "m3",
    "current_price": {
      "price_per_unit": "3.5000",
      "effective_from": "2025-01-01",
      "effective_to": null
    },
    "is_active": true
  }
}
```

PUT /config/water/{id}

Updates the water measurement unit. Does not affect pricing. To change the water price, use the Price History endpoint (POST /prices).

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Request Body (JSON)

```
{
  "unit": "litres"
}
```

Note: Changing the unit does not convert existing records. Old entries retain the unit they were recorded with via unit_at_entry.

6. Fertilizer Registry

Fertilizers are reference products with fixed NPK composition percentages. When a technician records a fertilization operation, the system snapshots the current NPK percentages and price at the time of entry so that later edits to the fertilizer record do not corrupt historical data.

GET /fertilizers

Lists all registered fertilizer products with their NPK composition and current price.

Auth: Bearer token | **Roles:** technician, manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
is_active	boolean	No	Default: true
search	string	No	Search by name

Response Body

```
{
  "data": [
    {
      "id": "f7g8...",
      "name": "NPK 15-15-15",
      "unit": "kg",
      "n_percent": "15.00",
      "p_percent": "15.00",
      "k_percent": "15.00",
      "current_price": { "price_per_unit": "12.0000", "effective_from": "2025-01-01" },
      "is_active": true
    }
  ]
}
```

POST /fertilizers

Registers a new fertilizer product. NPK percentages are stored as decimals (0-100). An initial price must be set via a follow-up call to POST /prices.

Auth: Bearer token | **Roles:** admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "name": "Ammonium Sulphate",
  "unit": "kg",
  "n_percent": "21.00",
  "p_percent": "0.00",
  "k_percent": "0.00"
}
```

PUT /fertilizers/{id}

Updates fertilizer attributes. NPK percentage changes apply only to future operations; historical entries retain their snapshot values.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Request Body (JSON)

```
{
  "name": "Ammonium Sulphate (Granular)",
  "n_percent": "21.00"
}
```

DELETE /fertilizers/{id}

Soft-deletes the fertilizer. It will no longer appear in the selection dropdown but remains in historical reports.

Auth: Bearer token | **Roles:** admin | **Success:** 204 No Content

7. Pesticide Registry

Pesticide products store their chemical composition as free text for traceability purposes. As with fertilizers, pricing is managed via the Price History system.

GET /pesticides

Lists registered pesticide products.

Auth: Bearer token | **Roles:** technician, manager, admin | **Success:** 200 OK

Response Body

```
{
  "data": [
    {
      "id": "h9i0...",
      "name": "Mancozeb 80 WP",
      "unit": "kg",
      "chemical_composition": "Mancozeb 80% (Dithiocarbamate)",
      "current_price": { "price_per_unit": "85.0000", "effective_from": "2025-02-01" },
      "is_active": true
    }
  ]
}
```

POST /pesticides

Registers a new pesticide product.

Auth: Bearer token | **Roles:** admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "name": "Lambda-Cyhalothrin 5 EC",
  "unit": "L",
  "chemical_composition": "Lambda-cyhalothrin 50g/L (Pyrethroid)"
}
```

PUT /pesticides/{id}

Updates pesticide attributes.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Request Body (JSON)

```
{
  "chemical_composition": "Lambda-cyhalothrin 50g/L (Synthetic Pyrethroid, Group 3A)"
}
```

DELETE /pesticides/{id}

Soft-deletes the pesticide product.

Auth: Bearer token | **Roles:** admin | **Success:** 204 No Content

8. Labour Rate Configuration

The labour rate is a global daily rate applied to harvest cost calculations (workers x days x daily rate). Like all pricing, the rate is managed through the Price History system to maintain full historical traceability.

GET /config/labor

Returns the current labour configuration with the active daily rate.

Auth: Bearer token | **Roles:** manager (read-only), admin | **Success:** 200 OK

Response Body

```
{
  "data": {
    "id": "j1k2...",
    "current_rate": {
      "price_per_unit": "150.0000",
      "effective_from": "2025-01-01",
      "effective_to": null
    },
    "is_active": true
  }
}
```

Note: To update the daily rate, use POST /prices with entity_type=labor.

9. Price History

The price history system is the single source of truth for all resource costs. Prices are append-only — when a price changes, a new record is inserted with the new effective date, and the previous record's `effective_to` is automatically set to the day before. This ensures that historical operations always reference the price that was active at the time they were recorded.

9.1 Supported Entity Types

Entity Type	Description	entity_id	Notes
water	Water unit price	NULL (global)	Price per m³, litre, etc.
fertilizer	Fertilizer product price	fertilizer UUID	Price per kg, L, etc.
pesticide	Pesticide product price	pesticide UUID	Price per kg, L, etc.
labor	Daily worker rate	NULL (global)	Rate per worker per day

POST /prices

Sets a new price for a resource. The system automatically closes the previous active price by setting its `effective_to` to the day before the new `effective_from`. If a price already exists for the same entity and overlapping date range, a 409 Conflict is returned.

Auth: Bearer token | **Roles:** admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "entity_type": "fertilizer",
  "entity_id": "f7g8...",
  "price_per_unit": "14.5000",
  "effective_from": "2025-04-01"
}
```

Response Body

```
{
  "data": {
    "id": "m3n4...",
    "entity_type": "fertilizer",
    "entity_id": "f7g8...",
    "price_per_unit": "14.5000",
    "effective_from": "2025-04-01",
    "effective_to": null,
    "created_at": "2025-03-28T10:00:00Z"
  }
}
```

Error Responses

Code	Description
409	Price already exists for this period

Code	Description
422	effective_from cannot be in the past

Note: For water and labor, omit entity_id (or pass null). For fertilizer and pesticide, entity_id is required and must reference an active product.

GET /prices

Queries the price timeline for a given entity. Returns all price records ordered by effective date, allowing the client to display a full price history.

Auth: Bearer token | **Roles:** manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
entity_type	string	Yes	water, fertilizer, pesticide, labor
entity_id	uuid	No	Required for fertilizer/pesticide
effective_at	date	No	Returns the single price active on this date

Response Body

```
{
  "data": [
    {
      "price_per_unit": "14.5000",
      "effective_from": "2025-04-01",
      "effective_to": null
    },
    {
      "price_per_unit": "12.0000",
      "effective_from": "2025-01-01",
      "effective_to": "2025-03-31"
    },
    {
      "price_per_unit": "10.5000",
      "effective_from": "2024-06-01",
      "effective_to": "2024-12-31"
    }
  ]
}
```

GET /prices/resolve

Returns the exact price that was (or would be) active for a given entity on a specific date. Used internally by the operation recording endpoints to snapshot price_at_entry.

Auth: Bearer token | **Roles:** technician, manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
entity_type	string	Yes	water, fertilizer, pesticide, labor
entity_id	uuid	No	Required for fertilizer/pesticide
date	date	Yes	The date to resolve the price for

Response Body

```
{
  "data": {
    "price_per_unit": "12.0000",
    "effective_from": "2025-01-01",
    "effective_to": "2025-03-31"
  }
}
```

Error Responses

Code	Description
404	No price defined for this entity on the given date

10. Irrigation Operations

Records water applied to a plot. The system resolves the current water price from price history and stores it as `price_at_entry` alongside the water unit used.

Request Fields

Field	Type	Required	Description
plot_id	uuid	Yes	Target plot
operation_date	date	Yes	Date of irrigation
water_quantity	decimal	Yes	Amount of water applied

GET /operations/irrigation

Lists irrigation operations with pagination. Filterable by plot and date range.

Auth: Bearer token | **Roles:** technician (own), manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
plot_id	uuid	No	Filter by plot
date_from	date	No	Start of date range
date_to	date	No	End of date range
page	integer	No	Page number

POST /operations/irrigation

Records a new irrigation operation. Price and configuration values are resolved and snapshotted server-side.

Auth: Bearer token | **Roles:** technician, admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "plot_id": "d4a2...",
  "operation_date": "2025-04-10",
  "water_quantity": "45.50"
}
```

Response Body

```
{
  "data": {
    "id": "01p2...",
    "plot_id": "d4a2...",
    "operation_date": "2025-04-10",
    "water_quantity": "45.50",
    "unit_at_entry": "m3",
    "price_at_entry": "3.5000",
  }
}
```

```
"created_by": "9c7f...",
"created_at": "2025-04-10T14:22:00Z"
}
}
```

Error Responses

Code	Description
404	Plot or resource not found
422	Validation failed

Note: The client does not send price or unit. These are resolved server-side from configuration and price history at the time of submission.

GET /operations/irrigation/{id}

Returns a single irrigation operation with full details including snapshotted prices and related entity names.

Auth: Bearer token | **Roles:** technician (own), manager, admin | **Success:** 200 OK

PUT /operations/irrigation/{id}

Updates a irrigation operation. Only the creating user or an admin can edit. Price snapshots are recalculated if the operation date changes.

Auth: Bearer token | **Roles:** technician (own), admin | **Success:** 200 OK

Note: Editing is only permitted within 48 hours of creation. After that, only admins can modify.

DELETE /operations/irrigation/{id}

Soft-deletes the irrigation operation.

Auth: Bearer token | **Roles:** admin | **Success:** 204 No Content

11. Fertilization Operations

Records a fertilizer application. The server resolves the fertilizer's NPK percentages and price, snapshotting them as `n_at_entry`, `p_at_entry`, `k_at_entry`, and `price_at_entry`.

Request Fields

Field	Type	Required	Description
plot_id	uuid	Yes	Target plot
fertilizer_id	uuid	Yes	Fertilizer product
operation_date	date	Yes	Date of application
quantity_applied	decimal	Yes	Amount applied

GET /operations/fertilization

Lists fertilization operations with pagination. Filterable by plot and date range.

Auth: Bearer token | **Roles:** technician (own), manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
plot_id	uuid	No	Filter by plot
date_from	date	No	Start of date range
date_to	date	No	End of date range
page	integer	No	Page number

POST /operations/fertilization

Records a new fertilization operation. Price and configuration values are resolved and snapshotted server-side.

Auth: Bearer token | **Roles:** technician, admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "plot_id": "d4a2...",
  "fertilizer_id": "f7g8...",
  "operation_date": "2025-04-10",
  "quantity_applied": "120.00"
}
```

Response Body

```
{
  "data": {
    "id": "q3r4...",
    "plot_id": "d4a2...",
    "fertilizer_id": "f7g8...",

```

```

    "operation_date": "2025-04-10",
    "quantity_applied": "120.00",
    "n_at_entry": "15.00",
    "p_at_entry": "15.00",
    "k_at_entry": "15.00",
    "price_at_entry": "12.0000",
    "created_at": "2025-04-10T14:30:00Z"
  }
}

```

Error Responses

Code	Description
404	Plot or resource not found
422	Validation failed

Note: NPK values are computed server-side. The client only submits the product and quantity.

GET /operations/fertilization/{id}

Returns a single fertilization operation with full details including snapshotted prices and related entity names.

Auth: Bearer token | **Roles:** technician (own), manager, admin | **Success:** 200 OK

PUT /operations/fertilization/{id}

Updates a fertilization operation. Only the creating user or an admin can edit. Price snapshots are recalculated if the operation date changes.

Auth: Bearer token | **Roles:** technician (own), admin | **Success:** 200 OK

Note: Editing is only permitted within 48 hours of creation. After that, only admins can modify.

DELETE /operations/fertilization/{id}

Soft-deletes the fertilization operation.

Auth: Bearer token | **Roles:** admin | **Success:** 204 No Content

12. Phytosanitary Operations

Records a pesticide treatment. Includes free-text fields for target pest and remarks to support traceability and compliance auditing.

Request Fields

Field	Type	Required	Description
plot_id	uuid	Yes	Target plot
pesticide_id	uuid	Yes	Pesticide product
operation_date	date	Yes	Date of treatment
quantity_applied	decimal	Yes	Amount applied
target_pest	string	No	Pest being treated
remarks	string	No	Observations

GET /operations/phytosanitary

Lists phytosanitary operations with pagination. Filterable by plot and date range.

Auth: Bearer token | **Roles:** technician (own), manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
plot_id	uuid	No	Filter by plot
date_from	date	No	Start of date range
date_to	date	No	End of date range
page	integer	No	Page number

POST /operations/phytosanitary

Records a new phytosanitary operation. Price and configuration values are resolved and snapshotted server-side.

Auth: Bearer token | **Roles:** technician, admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "plot_id": "d4a2...",
  "pesticide_id": "h9i0...",
  "operation_date": "2025-04-10",
  "quantity_applied": "2.50",
  "target_pest": "Ceratitis capitata (Mediterranean fruit fly)",
  "remarks": "Applied at dawn, wind <10km/h"
}
```

Response Body

```
{
  "data": {
    "id": "s5t6...",
    "plot_id": "d4a2...",
    "pesticide_id": "h9i0...",
    "operation_date": "2025-04-10",
    "quantity_applied": "2.50",
    "target_pest": "Ceratitis capitata (Mediterranean fruit fly)",
    "remarks": "Applied at dawn, wind <10km/h",
    "price_at_entry": "85.0000",
    "created_at": "2025-04-10T06:15:00Z"
  }
}
```

Error Responses

Code	Description
404	Plot or resource not found
422	Validation failed

Note: Chemical composition is not sent by the client; it is stored on the pesticide record and included when querying operation details.

GET /operations/phytosanitary/{id}

Returns a single phytosanitary operation with full details including snapshotted prices and related entity names.

Auth: Bearer token | **Roles:** technician (own), manager, admin | **Success:** 200 OK

PUT /operations/phytosanitary/{id}

Updates a phytosanitary operation. Only the creating user or an admin can edit. Price snapshots are recalculated if the operation date changes.

Auth: Bearer token | **Roles:** technician (own), admin | **Success:** 200 OK

Note: Editing is only permitted within 48 hours of creation. After that, only admins can modify.

DELETE /operations/phytosanitary/{id}

Soft-deletes the phytosanitary operation.

Auth: Bearer token | **Roles:** admin | **Success:** 204 No Content

13. Harvest Operations

Records a harvest event with labour and yield data. The daily labour rate is resolved from price history and stored as `daily_rate_at_entry`.

Request Fields

Field	Type	Required	Description
plot_id	uuid	Yes	Target plot
operation_date	date	Yes	Date of harvest
num_workers	integer	Yes	Worker count
days_worked	decimal	Yes	Duration in days
quantity_harvested	decimal	Yes	Yield amount

GET /operations/harvest

Lists harvest operations with pagination. Filterable by plot and date range.

Auth: Bearer token | **Roles:** technician (own), manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
plot_id	uuid	No	Filter by plot
date_from	date	No	Start of date range
date_to	date	No	End of date range
page	integer	No	Page number

POST /operations/harvest

Records a new harvest operation. Price and configuration values are resolved and snapshotted server-side.

Auth: Bearer token | **Roles:** technician, admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "plot_id": "d4a2...",
  "operation_date": "2025-04-10",
  "num_workers": 8,
  "days_worked": "3.00",
  "quantity_harvested": "2400.00"
}
```

Response Body

```
{
  "data": {
    "id": "u7v8...",
  }
}
```

```

    "plot_id": "d4a2...",
    "operation_date": "2025-04-10",
    "num_workers": 8,
    "days_worked": "3.00",
    "quantity_harvested": "2400.00",
    "daily_rate_at_entry": "150.0000",
    "created_at": "2025-04-10T17:00:00Z"
  }
}

```

Error Responses

Code	Description
404	Plot or resource not found
422	Validation failed

Note: Labour cost = num_workers x days_worked x daily_rate_at_entry. This calculation is performed at reporting time, not at entry.

GET /operations/harvest/{id}

Returns a single harvest operation with full details including snapshotted prices and related entity names.

Auth: Bearer token | **Roles:** technician (own), manager, admin | **Success:** 200 OK

PUT /operations/harvest/{id}

Updates a harvest operation. Only the creating user or an admin can edit. Price snapshots are recalculated if the operation date changes.

Auth: Bearer token | **Roles:** technician (own), admin | **Success:** 200 OK

Note: Editing is only permitted within 48 hours of creation. After that, only admins can modify.

DELETE /operations/harvest/{id}

Soft-deletes the harvest operation.

Auth: Bearer token | **Roles:** admin | **Success:** 204 No Content

14. Offline Sync (Postings)

The postings system is the bridge between offline mobile submissions and the authoritative server database. Every form submission from the mobile PWA — whether online or offline — first creates a posting record. The server then processes the posting, creates the corresponding operation record, and links them via `posting_id`.

The `client_id` field acts as an idempotency key: if the same `client_id` is submitted twice (which happens when the device retries after a network timeout), the server returns the existing result rather than creating a duplicate.

Posting Lifecycle

Status	Description
pending	Received by server, not yet processed
synced	Successfully processed; operation record created
failed	Processing failed; see <code>error_message</code> for details
conflict	Duplicate <code>client_id</code> detected; original record returned

POST /postings

Submits a single operation from the mobile device. The payload contains the same fields as the corresponding operation endpoint. The server processes synchronously and returns the created operation if successful.

Auth: Bearer token | **Roles:** technician, admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "client_id": "device-a1b2-20250410-001",
  "operation_type": "irrigation",
  "payload": {
    "plot_id": "d4a2...",
    "operation_date": "2025-04-10",
    "water_quantity": "45.50"
  },
  "submitted_at": "2025-04-10T14:22:00+01:00",
  "device_info": {
    "platform": "Android 14",
    "app_version": "1.2.0",
    "device_model": "Samsung A54"
  }
}
```

Response Body

```
{
  "data": {
```

```
    "id": "w9x0...",
    "client_id": "device-a1b2-20250410-001",
    "status": "syncd",
    "operation_type": "irrigation",
    "syncd_at": "2025-04-10T14:22:01Z",
    "operation": {
      "id": "o1p2...",
      "water_quantity": "45.50",
      "price_at_entry": "3.5000"
    }
  }
}
```

Error Responses

Code	Description
409	client_id already exists; returns original posting
422	Payload validation failed

POST /postings/batch

Submits multiple operations in a single request. Each posting is processed independently; partial success is possible. The response includes per-item status.

Auth: Bearer token | **Roles:** technician, admin | **Success:** 207 Multi-Status

Request Body (JSON)

```
{
  "postings": [
    { "client_id": "...", "operation_type": "irrigation", "payload": {...}, "submitted_at": "..."},
    { "client_id": "...", "operation_type": "fertilization", "payload": {...}, "submitted_at": "..."}
  ]
}
```

Response Body

```
{
  "data": {
    "processed": 2,
    "succeeded": 1,
    "failed": 1,
    "results": [
      { "client_id": "...", "status": "syncd", "operation": {...} },
      { "client_id": "...", "status": "failed", "error_message": "Plot not found" }
    ]
  }
}
```

Note: Maximum batch size is 50 postings per request.

GET /postings

Lists postings for the authenticated user, filterable by status. Used by the mobile sync status screen.

Auth: Bearer token | **Roles:** technician (own), admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
status	string	No	Filter: pending, synced, failed, conflict
date_from	date	No	Filter by submitted_at

Response Body

```
{
  "data": [
    {
      "id": "w9x0...",
      "client_id": "device-a1b2-20250410-001",
      "operation_type": "irrigation",
      "status": "syncd",
      "submitted_at": "2025-04-10T14:22:00+01:00",
      "syncd_at": "2025-04-10T14:22:01Z"
    }
  ]
}
```

POST /postings/{id}/retry

Retries processing a failed posting. The server re-validates the payload and attempts to create the operation record again.

Auth: Bearer token | **Roles:** technician (own), admin | **Success:** 200 OK

Error Responses

Code	Description
400	Posting is not in failed status
422	Payload still invalid

15. Reporting & Analytics

All report endpoints are read-only and compute aggregations on the fly from operation records. Reports respect the `price_at_entry` and NPK snapshots stored on each operation, ensuring that cost and nutrient calculations remain accurate regardless of subsequent configuration changes.

Every report endpoint accepts `plot_id` (optional, defaults to all plots) and `date_from` / `date_to` (optional, defaults to the current season based on the plot's `season_start_date`).

GET /reports/irrigation

Returns water usage aggregated by month and cumulative since season start. Values are expressed both as absolute quantities and per-hectare using the plot's surface area.

Auth: Bearer token | **Roles:** manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
plot_id	uuid	No	Specific plot or all
date_from	date	No	Period start
date_to	date	No	Period end
group_by	string	No	month (default), week, day

Response Body

```
{
  "data": {
    "plot": { "id": "d4a2...", "name": "Bloc A", "surface_area_ha": "12.5000" },
    "period": { "from": "2024-10-01", "to": "2025-04-10" },
    "monthly": [
      { "month": "2024-10", "quantity": "580.00", "quantity_per_ha": "46.40" },
      { "month": "2024-11", "quantity": "420.00", "quantity_per_ha": "33.60" }
    ],
    "cumulative": { "quantity": "2850.00", "quantity_per_ha": "228.00" }
  }
}
```

GET /reports/fertilization

Returns NPK nutrient load by month and cumulative, calculated from the snapshotted `n/p/k_at_entry` values on each fertilization operation. All values are per hectare.

Auth: Bearer token | **Roles:** manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
plot_id	uuid	No	Specific plot or all
date_from	date	No	Period start

Parameter	Type	Required	Description
date_to	date	No	Period end

Response Body

```
{
  "data": {
    "plot": { "id": "d4a2...", "name": "Bloc A", "surface_area_ha": "12.5000" },
    "monthly": [
      {
        "month": "2024-11",
        "n_units_per_ha": "14.40",
        "p_units_per_ha": "14.40",
        "k_units_per_ha": "14.40"
      }
    ],
    "cumulative": {
      "n_units_per_ha": "38.20",
      "p_units_per_ha": "32.10",
      "k_units_per_ha": "28.50"
    }
  }
}
```

Note: NPK units per hectare = (quantity_applied x n/p/k_at_entry / 100) / surface_area_ha

GET /reports/phytosanitary

Returns a detailed treatment log per plot, designed for regulatory traceability. Each row includes the pesticide name, chemical composition (from the pesticide record), target pest, and remarks.

Auth: Bearer token | **Roles:** manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
plot_id	uuid	No	Specific plot or all
search	string	No	Filter across pesticide name, pest, or remarks

Response Body

```
{
  "data": [
    {
      "operation_date": "2025-04-10",
      "pesticide": {
        "name": "Mancozeb 80 WP",
        "chemical_composition": "Mancozeb 80% (Dithiocarbamate)"
      },
      "quantity_applied": "2.50",
      "target_pest": "Ceratitis capitata",
    }
  ]
}
```

```

    "remarks": "Applied at dawn",
    "recorded_by": "Ahmed Benali"
  }
]
}

```

GET /reports/harvest

Returns harvest logs per plot with labour and yield details.

Auth: Bearer token | **Roles:** manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
plot_id	uuid	No	Specific plot or all
date_from	date	No	Period start
date_to	date	No	Period end

Response Body

```

{
  "data": [
    {
      "operation_date": "2025-04-10",
      "num_workers": 8,
      "days_worked": "3.00",
      "quantity_harvested": "2400.00",
      "labour_cost": "3600.0000",
      "recorded_by": "Ahmed Benali"
    }
  ]
}

```

GET /reports/production-cost

Aggregates all costs per plot for a given period: irrigation, fertilization, phytosanitary, and harvest labour. Each cost line uses the price_at_entry snapshot from the individual operations.

Auth: Bearer token | **Roles:** manager, admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
plot_id	uuid	No	Specific plot or all
date_from	date	No	Period start
date_to	date	No	Period end

Response Body

```

{
  "data": [

```

```

{
  "plot": { "id": "d4a2...", "name": "Bloc A", "surface_area_ha": "12.5000" },
  "irrigation_cost": "9975.00",
  "fertilization_cost": "18720.00",
  "phytosanitary_cost": "4250.00",
  "harvest_cost": "14400.00",
  "total_cost": "47345.00",
  "cost_per_ha": "3787.60"
}
],
"summary": {
  "total_irrigation": "9975.00",
  "total_fertilization": "18720.00",
  "total_phytosanitary": "4250.00",
  "total_harvest": "14400.00",
  "grand_total": "47345.00"
}
}

```

Note: Irrigation cost = SUM(water_quantity x price_at_entry). Fertilization cost = SUM(quantity_applied x price_at_entry). Phytosanitary cost = SUM(quantity_applied x price_at_entry). Harvest cost = SUM(num_workers x days_worked x daily_rate_at_entry).

16. Custom Fields (EAV)

The Entity-Attribute-Value system allows administrators to extend any entity (plots, fertilizers, pesticides) with additional fields without schema migrations. Custom fields are defined once and then values are attached to individual entity records.

GET /custom-fields

Lists all custom field definitions, optionally filtered by entity type.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
entity_type	string	No	Filter: plot, fertilizer, pesticide

Response Body

```
{
  "data": [
    {
      "id": "y1z2...",
      "entity_type": "plot",
      "field_name": "soil_ph",
      "field_label": "Soil pH",
      "field_type": "number",
      "is_required": false,
      "display_order": 1
    }
  ]
}
```

POST /custom-fields

Defines a new custom field. For select type fields, provide the options array.

Auth: Bearer token | **Roles:** admin | **Success:** 201 Created

Request Body (JSON)

```
{
  "entity_type": "plot",
  "field_name": "irrigation_method",
  "field_label": "Irrigation Method",
  "field_type": "select",
  "is_required": false,
  "select_options": ["drip", "flood", "sprinkler", "furrow"],
  "display_order": 2
}
```

PUT /custom-fields/{id}/values/{entity_id}

Sets or updates a custom field value for a specific entity instance.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Request Body (JSON)

```
{
  "value": "drip"
}
```

Note: The server determines which value column to use (value_text, value_number, value_date, value_bool) based on the field_type definition.

17. System Logs

The logging system captures authentication events, CRUD operations, sync activity, and system errors. Logs are append-only and cannot be modified or deleted through the API. Retention policy: 90 days for info/debug, 1 year for warn/error.

GET /logs

Queries system logs with filtering. Results are ordered by creation time, most recent first.

Auth: Bearer token | **Roles:** admin | **Success:** 200 OK

Query Parameters

Parameter	Type	Required	Description
level	string	No	Filter: info, warn, error, debug
category	string	No	Filter: auth, sync, crud, system, price
user_id	uuid	No	Filter by acting user
entity_type	string	No	Filter by affected entity type
entity_id	uuid	No	Filter by affected entity
date_from	datetime	No	Start of time range
date_to	datetime	No	End of time range
page	integer	No	Page number
per_page	integer	No	Results per page (max 100)

Response Body

```
{
  "data": [
    {
      "id": 15847,
      "level": "info",
      "category": "crud",
      "action": "plot.created",
      "entity_type": "plot",
      "entity_id": "d4a2...",
      "details": { "name": "Bloc B - Olives" },
      "user_id": "9c7f...",
      "ip_address": "196.200.130.45",
      "created_at": "2025-04-10T14:30:00Z"
    }
  ],
  "meta": { "current_page": 1, "per_page": 50, "total": 15847 }
}
```

18. Internationalisation

The API supports French (fr) and English (en). The client communicates its preferred language through two mechanisms:

Mechanism	When Applied	Example
Accept-Language header	Sent with each request	Accept-Language: fr
User preference	Stored in user profile	preferred_lang field on /users
Fallback	When neither is set	French (fr) is the default

Server-side error messages, validation messages, and any text content in responses are returned in the requested language. Numeric formatting (decimal separator, thousands grouping) follows the locale: French uses comma as decimal separator, English uses period.

Date formats in API responses always use ISO 8601 regardless of locale. Date formatting for display is the client's responsibility.

End of Document

Agri-Sync REST API Reference v1.0.0 — Agri-Sync Engineering

For questions: engineering@agrisync.com